Class 14: Project Proposal and Delivery Planning

Data-driven Systems Engineering

Agenda

- Why planning matters in ML projects
- Project summary and scope
- Deliverables and milestones
- Work Breakdown Structure and sprint or Gantt planning
- Definition of Ready and Definition of Done
- Risks and governance for ML projects

Planning is part of the engineering work

Learning goals

- Turn a project idea into scope, milestones, and deliverables.
- Define what will be built first and what can wait.
- Make project status visible through artifacts, not only verbal updates.
- Connect proposal quality to later testing, deployment, and monitoring work.

Course Map and Running Cases

Problem → Data → Features → Model → Interface → Delivery → Monitor → Evolution

Today in the course

Focus: Project planning as lifecycle design.
Bridge: This class turns the course concepts into an executable project plan with milestones, deliverables, and evidence of progress.

Fraud case

In the fraud case, planning means sequencing data work, baseline modeling, service exposure, deployment checks, and monitoring criteria.

Pasteurization case

In the pasteurization case, planning means sequencing sensor understanding, forecasting or alerting logic, dashboard integration, and drift response design.

Course thread: The course thread is that a coherent project proposal should already reflect the full lifecycle, not only the modeling step.

Why planning matters in ML projects

- ML systems evolve through data, code, models, interfaces, validation, and operations.
- More roles means more handoffs, which means more need for shared artifacts.
- A weak proposal creates downstream confusion in requirements, evaluation, and release planning.

Practical claim

Planning does not remove uncertainty, but it makes uncertainty visible enough to manage.

A project summary defines boundaries

- What problem is being solved?
- Who uses or is affected by the result?
- What is in scope for this project?
- What is intentionally out of scope?
- What evidence will convince stakeholders that the project is useful?

What a strong project summary should contain

- Problem statement and target users.
- Data sources and expected constraints.
- Expected system output: prediction, forecast, alert, or recommendation.
- Success criteria tied to technical and user outcomes.
- Risks, assumptions, and dependencies.

Objectives should be specific enough to evaluate

Weak objective

Build an ML system for quality control.

Stronger objective

Detect abnormal process trajectories from synthetic pasteurization sensor data and expose alerts through a dashboard with traceable event logs.

- Strong objectives are specific, measurable, relevant, and bounded enough to review.

Milestones and WBS

- Proposal approval
- Data pipeline MVP
- Baseline model
- Deployable service or dashboard
- Monitoring active
- Final demonstration

WBS reminder

Break work into outputs that can be reviewed, not vague activity labels.

The WBS is about outputs, not vague activity

- Good work packages end in something reviewable: cleaned dataset, baseline notebook, API endpoint, dashboard page, monitoring report.
- The 100% rule helps ensure the WBS covers the whole scope without duplication.
- If a task cannot be evaluated by an external reviewer, it is often too vague.

Suggested semester deliverables

1. Data understanding and requirement brief.
2. Baseline model and evaluation report.
3. Packaged service or dashboard.
4. CI/CD or reproducibility automation.
5. Monitoring or drift analysis plan.

Deliverables should map to the lifecycle

Deliverable	Why it matters
Data pipeline artifact	Makes inputs reproducible and inspectable.
Baseline model	Creates a reference point for later improvement.
Deployable interface	Proves the model can be consumed by users or systems.
Automation artifact	Reduces manual, unrepeatable work.
Monitoring plan	Shows how the system remains trustworthy after release.

Sprint and timeline thinking

- Sprint 0: environment, repo structure, and scope.
- Sprint 1: data understanding and requirements.
- Sprint 2: baseline modeling.
- Sprint 3: serving and integration.
- Sprint 4: monitoring and final hardening.

Milestones reduce risk when they represent decisions

- Proposal approval means the team has a coherent scope.
- Data pipeline MVP means later modeling is grounded in real inputs.
- Baseline model means performance can be compared instead of guessed.
- Deployable service means integration risk is no longer abstract.
- Monitoring active means the project is thinking beyond the demo.

A more suitable proposal example

Project

Forecast temperature and detect abnormal state transitions in a synthetic pasteurization line.

- Data source: Flask stream from ‘pasteurization/serving.py‘.
- MVP: predict next temperature value and visualize drift alerts.
- Stretch goal: state-aware forecasting and alert triage.
- Main risks: synthetic realism, label scarcity, and operator-facing thresholds.

Another project pattern students can defend

Project

Fraud scoring as a service for transaction review support.

- Data source: credit-card fraud dataset and engineered features.
- MVP: classify transactions and expose a scoring endpoint.
- Stretch goal: threshold tuning plus analyst-facing dashboard.
- Risks: class imbalance, false-positive cost, and drift after deployment.

DoR and DoD for student teams

Definition of Ready

- Requirement is written
- Data source identified
- Acceptance criteria drafted

Definition of Done

- Code runs reproducibly
- Result is documented
- Test or demo evidence exists

A lightweight risk register helps project control

Risk	Example response
Data access or quality risk	Define fallback datasets and validation checks.
Model performance risk	Commit to a baseline and threshold review.
Integration risk	Deliver an API or dashboard slice early.
Schedule risk	Separate must-have deliverables from stretch goals.
Governance risk	Document assumptions, versions, and monitoring triggers.

What a strong proposal presentation should show

- The problem and why it matters.
- The system boundary and main users.
- The data source, constraints, and risks.
- The milestone sequence and what each one proves.
- The evidence that the project is becoming an engineered system, not only a notebook.

From This Class to the Next

What stays with us

- A strong proposal defines the problem, the system boundary, the evidence plan, and the main risks.
- Milestones should demonstrate engineering progress, not only analytical activity.
- Good project planning anticipates delivery, monitoring, and change from the start.

Project use

This helps student teams present a believable path from data and baseline models to interfaces, deployment, and operational follow-up.

Next connection

Class 15 closes the sequence with monitoring and drift response, showing how project plans must extend beyond first deployment.

Course thread: Project planning becomes complete only when it includes how the system will be observed, corrected, and improved after release.

Papers and materials

Papers and materials

Sculley et al. (2015), Hidden Technical Debt; Designing Machine Learning Systems; Project planning with epics and stories; Minimum Viable Product guide; Critical path and schedule planning basics.